

# Database Normalization Report

Bank Management System — Gebar Commercial Bank

AASTU | Software Engineering | Database Systems | Group 5

This document demonstrates the normalization of the Bank Management System database for Gebar Commercial Bank, progressing through First Normal Form (1NF), Second Normal Form (2NF), Third Normal Form (3NF), and Boyce-Codd Normal Form (BCNF). Each step removes data redundancy and update anomalies, producing a clean, dependency-preserving relational schema.

| Normal Form | Key Rule                                | Status    |
|-------------|-----------------------------------------|-----------|
| 1NF         | Atomic values, no repeating groups      | SATISFIED |
| 2NF         | No partial dependencies on composite PK | SATISFIED |
| 3NF         | No transitive dependencies              | SATISFIED |
| BCNF        | Every determinant is a candidate key    | SATISFIED |

## 0 Step 0 — Unnormalized Form (UNF)

Before normalization, all banking data lives in a single flat relation. This creates repeating groups, update anomalies, and significant data redundancy.

| customer_id | customer_name | account_no       | account_balance        | branch_id | branch_name | employee_id | employee_position | loan_id        | loan_type              | transaction_id | transaction_amount    | user_id | user_role | log_id | table_name | payment_id    | payment_date |
|-------------|---------------|------------------|------------------------|-----------|-------------|-------------|-------------------|----------------|------------------------|----------------|-----------------------|---------|-----------|--------|------------|---------------|--------------|
| C001        | Abebe Bekele  | A1001<br>, A1002 | 5000.00<br>, 12,000.00 | B001      | Main Branch | E001        | manager           | L001           | Home                   | T001,<br>T002  | 15000.00,<br>20000.00 | U001    | manager   | AL001  | customer,  | P001,<br>P004 | 2026-03-15   |
| C002        | Hana Tesfaye  | A1003            | 8000.00                | B002      | Bole Branch | E002        | cashier           | L002<br>, L003 | Business,<br>Education | T003           | 20000.00              | U002    | customer  | AL002  | account    | P002          | 2026-04-15   |

Problems: customer\_name and branch repeat for every row. Accounts, loans, and transactions are stored as comma-separated lists — violating atomicity. Deleting the last transaction for a customer would lose customer data entirely.

## 1 Step 1 — First Normal Form (1NF)

Rule: Each column must contain atomic (indivisible) values. No repeating groups or multi-valued attributes.

### Actions taken:

1. Assigned a primary key to every relation.
2. Ensured every attribute holds a single, indivisible value (e.g., phone is one number per row, not a list).
3. Eliminated all comma-separated multi-values by expanding into separate rows.

### Resulting table after 1NF:

| customer_id | customer_name | account_no | account_balance | branch_id | branch_name | employee_id | employee_position | loan_id | loan_type | transaction_id | transaction_amount | user_id | user_role | log_id | table_name | payment_id | payment_date |
|-------------|---------------|------------|-----------------|-----------|-------------|-------------|-------------------|---------|-----------|----------------|--------------------|---------|-----------|--------|------------|------------|--------------|
| C001        | Abebe Bekele  | A1001      | 5000.00, 1      | B001      | Main Branch | E001        | manager           | L001    | Home      | T001           | 15000.00           | U001    | manager   | AL001  | customer,  | P001       | 2026-03-15   |
| C002        | Hana Tesfaye  | A1003      | 8000.00         | B002      | Bole Branch | E002        | cashier           | L002    | Business  | T003           | 20000.00           | U002    | customer  | AL002  | account    | P002       | 2026-04-15   |

All relations now contain only atomic values. 1NF is satisfied.

## 2 Step 2 — Second Normal Form (2NF)

Rule: Must be in 1NF, and every non-key attribute must be fully functionally dependent on the entire primary key (relevant when the PK is composite).

In this schema, all relations use single-column primary keys, so there are no composite PKs and therefore no partial dependencies can exist. Each non-key attribute depends fully on its table's single PK.

### Functional Dependencies identified:

| Determinant    | Dependent Attributes                                                                                                                                                                                |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| customer_id    | first_name, last_name, phone_number, registration_date, email, national_id, branch_id                                                                                                               |
| account_no     | customer_id, account_type, balance, open_date, status, branch_id                                                                                                                                    |
| loan_id        | customer_id, branch_id, loan_type, loan_amount, interest_rate, duration_months, approval_date, status, application_date, disbursement_date, monthly_installment, outstanding_balance, next_due_date |
| transaction_id | account_no, transaction_type, amount, transaction_date, description, reference_number, branch_id                                                                                                    |
| employee_id    | first_name, last_name, position, salary, email, phone_number, hire_date, branch_id                                                                                                                  |
| branch_id      | branch_name, address, phone_number, manager_id                                                                                                                                                      |
| payment_id     | payment_date, amount_paid, principal_paid, interest_paid, remaining_balance, employee_id, loan_id                                                                                                   |
| user_id        | user_name, password, role, status, employee_id, customer_id, last_login                                                                                                                             |
| log_id         | table_name, record_id, old_values, new_values action, changed_at, description, employee_id                                                                                                          |

### Schema after 2NF (unchanged from 1NF):

| Relation      | Primary Key    | Foreign Keys           | Other Attributes                                                                             |
|---------------|----------------|------------------------|----------------------------------------------------------------------------------------------|
| Branch        | branch_id      | —                      | branch_name, address, phone_number, manager_id                                               |
| Customer      | customer_id    | branch_id              | first_name, last_name, phone_number, email, registration_date, national_id                   |
| Account       | account_no     | customer_id, branch_id | account_type, balance, open_date, status, application_date                                   |
| Loan          | loan_id        | customer_id, branch_id | loan_type, loan_amount, interest_rate, duration_months, start_date, status, application_date |
| Transaction   | transaction_id | account_no, branch_id  | transaction_type, amount, transaction_date, description, reference_number                    |
| Employee      | employee_id    | branch_id              | first_name, last_name, position, salary, email, phone_number, hire_date                      |
| Loan Payments | payment_id     | loan_id                | payment_date, amount_paid, principal_paid, interest_paid, remaining_balance                  |

|           |         |                             |                                                                                  |
|-----------|---------|-----------------------------|----------------------------------------------------------------------------------|
| Users     | user_id | employee_id,<br>customer_id | user_name, password, role, status, last_login                                    |
| Audit Log | log_id  | employee_id                 | table_name, record_id, old_values, new_values<br>action, changed_at, description |

**No partial dependencies exist. 2NF is satisfied.**

### 3 Step 3 — Third Normal Form (3NF)

Rule: Must be in 2NF, and no non-key attribute may transitively depend on the primary key through another non-key attribute (no dependency chains of the form  $PK \rightarrow X \rightarrow Y$  where  $X$  is not a candidate key). We check every relation for transitive dependencies of the form  $PK \rightarrow X \rightarrow Y$  where  $X$  is not a key. Analysis of each relation:

| Relation      | Dependency Check                                                                                                                                                                         | Verdict |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| Branch        | branch_id → branch_name, address, phone_number, manager_id. All attributes depend directly on the PK. No transitive chain.                                                               | OK      |
| Customer      | customer_id → branch_id is a FK reference, not a non-key attribute causing a transitive chain. Branch details live in a separate Branch table. No transitive dependency within Customer. | OK      |
| Account       | account_no → customer_id, branch_id are FK references. All remaining attributes (type, balance, status) depend directly on account_no. No transitive dependency.                         | OK      |
| Loan          | loan_id → customer_id, branch_id are FK references. Loan details (amount, rate, duration, status) all depend directly on loan_id. No transitive dependency.                              | OK      |
| Transaction   | transaction_id → account_no, branch_id are FK references. Amount, type, date, description all depend directly on transaction_id. No transitive dependency.                               | OK      |
| Employee      | employee_id → branch_id is a FK reference. Employee details (name, position, salary, hire_date) all depend directly on employee_id. No transitive dependency.                            | OK      |
| Loan Payments | payment_id → loan_id is a FK reference. Payment details(payment_date, amount_paid, principal_paid, interest_paid, remaining_balance) all depend directly on payment_id.                  | OK      |
| Users         | user_id → employee_id, customer_id are FK references. User details(user_name, password, role, status, last_login) all depend directly on user_id.                                        | OK      |
| Audit Log     | log_id → employee_id is a FK reference. Log details(table_name, record_id, old_values, new_values action, changed_at, description) all depend directly on log_id.                        | OK      |

## 4 Step 4 — Boyce-Codd Normal Form (BCNF)

**Rule:** For every non-trivial functional dependency  $X \rightarrow Y$ ,  $X$  must be a superkey (candidate key or superset of one).

BCNF is stricter than 3NF and catches lingering anomalies that 3NF can miss when there are overlapping candidate keys. Based on the system's strict architectural determinants, the analytical check updates as follows:

**Candidate Key Analysis:**

| Relation      | Candidate Key(s) | All Determinants Are Superkeys? |
|---------------|------------------|---------------------------------|
| Branch        | branch_id        | Yes                             |
| Customer      | customer_id      | Yes                             |
| Account       | account_no       | Yes                             |
| Loan          | loan_id          | Yes                             |
| Transaction   | transaction_id   | Yes                             |
| Employee      | employee_id      | Yes                             |
| Loan Payments | payment_id       | Yes                             |
| Users         | user_id          | Yes                             |
| Audit Log     | log_id           | Yes                             |

In each relation, the only determinants in non-trivial functional dependencies are the primary keys, which act as proper candidate keys. There are no BCNF violations.

# Complete Set of Functional Dependencies (Final Schema)

| Determinant    | Dependent Attributes                                                                                                                                                                               |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| customer_id    | first_name, last_name, phone_number, registration_date, email, national_id, branch_id                                                                                                              |
| account_no     | customer_id, account_type, balance, open_date, status, branch_id                                                                                                                                   |
| loan_id        | customer_id, branch_id, loan_type, loan_amount, interest_rate, duration_months, approval_date, status, application_date, disbrusment_date, monthly_installment, outstanding_balance, next_due_date |
| transaction_id | account_no, transaction_type, amount, transaction_date, description, reference_number, branch_id                                                                                                   |
| employee_id    | first_name, last_name, position, salary, email, phone_number, hire_date, branch_id                                                                                                                 |
| branch_id      | branch_name, address, phone_number, manager_id                                                                                                                                                     |
| payment_id     | payment_date, amount_paid, principal_paid, interest_paid, remaining_balance, employee_id, loan_id                                                                                                  |
| user_id        | user_name, password, role, status, employee_id, customer_id, last_login                                                                                                                            |
| log_id         | table_name, record_id, old_values, new_values action, changed_at, description, employee_id                                                                                                         |

All determinants are candidate keys. BCNF is satisfied.

## Final Normalized Schema

The final operational data dictionaries after achieving complete BCNF mapping. All tables are completely free from data redundancy, partial dependencies, transitive chains, or superkey overlap vulnerabilities.

### 1. BRANCH

| Column       | Type         | Key | Description              |
|--------------|--------------|-----|--------------------------|
| branch_id    | VARCHAR(10)  | PK  | Unique branch identifier |
| branch_name  | VARCHAR(100) |     | Official branch name     |
| address      | VARCHAR(200) |     | Branch physical address  |
| phone_number | VARCHAR(20)  |     | Branch contact number    |

|            |             |    |                     |
|------------|-------------|----|---------------------|
| manager_id | VARCHAR(20) | FK | References EMPLOYEE |
|------------|-------------|----|---------------------|

## 2. CUSTOMER

| Column            | Type         | Key | Description                    |
|-------------------|--------------|-----|--------------------------------|
| customer_id       | VARCHAR(15)  | PK  | Unique customer identifier     |
| first_name        | VARCHAR(50)  |     | Customer first name            |
| last_name         | VARCHAR(50)  |     | Customer last name             |
| phone_number      | VARCHAR(20)  |     | Contact phone number           |
| registration_date | DATE         |     | Date customer joined the bank  |
| email             | VARCHAR(100) |     | Email address                  |
| national_id       | VARCHAR(30)  |     | National identification number |
| branch_id         | VARCHAR(10)  | FK  | References BRANCH              |

## 3. ACCOUNT

| Column       | Type           | Key | Description                   |
|--------------|----------------|-----|-------------------------------|
| account_no   | VARCHAR(20)    | PK  | Unique account number         |
| customer_id  | VARCHAR(15)    | FK  | References CUSTOMER           |
| account_type | VARCHAR(20)    |     | Savings or Current            |
| balance      | DECIMAL(15, 2) |     | Current account balance (ETB) |
| open_date    | DATE           |     | Date account was opened       |
| status       | VARCHAR(20)    |     | Active / Inactive / Closed    |

|           |             |    |                   |
|-----------|-------------|----|-------------------|
| branch_id | VARCHAR(10) | FK | References BRANCH |
|-----------|-------------|----|-------------------|

4. LOAN

| Column              | Type           | Key | Description                         |
|---------------------|----------------|-----|-------------------------------------|
| loan_id             | VARCHAR(20)    | PK  | Unique loan identifier              |
| customer_id         | VARCHAR(15)    | FK  | References CUSTOMER                 |
| branch_id           | VARCHAR(10)    | FK  | References BRANCH                   |
| loan_type           | VARCHAR(50)    |     | Education, Business, Personal, etc. |
| loan_amount         | DECIMAL(15, 2) |     | Principal loan amount (ETB)         |
| interest_rate       | DECIMAL(5, 2)  |     | Annual interest rate (%)            |
| duration_months     | INT            |     | Loan duration in months             |
| approval_date       | DATE           |     | Date loan was officially approved   |
| status              | VARCHAR(20)    |     | Pending / Active / Closed           |
| application_date    | DATE           |     | Loan applied date                   |
| disbursement_date   | DATE           |     | Loan disbursement date              |
| monthly_installment | DECIMAL(15, 2) |     | Fixed monthly payment amount (ETB)  |
| outstanding_balance | DECIMAL(15, 2) |     | Remaining balance principal (ETB)   |
| next_due_date       | DATE           |     | Deadline for the next installment   |

5. TRANSACTION

| Column           | Type        | Key | Description                   |
|------------------|-------------|-----|-------------------------------|
| transaction_id   | VARCHAR(20) | PK  | Unique transaction identifier |
| account_no       | VARCHAR(20) | FK  | References ACCOUNT            |
| transaction_type | VARCHAR(20) |     | Deposit, Withdrawal, Transfer |



|                  |                |    |                                      |
|------------------|----------------|----|--------------------------------------|
| amount           | DECIMAL(15, 2) |    | Transaction amount (ETB)             |
| transaction_date | DATE           |    | Date of transaction                  |
| description      | VARCHAR(200)   |    | Optional transaction note            |
| reference_number | VARCHAR(50)    |    | External reference or receipt number |
| branch_id        | VARCHAR(10)    | FK | References BRANCH                    |

6. EMPLOYEE

| Column       | Type           | Key | Description                |
|--------------|----------------|-----|----------------------------|
| employee_id  | VARCHAR(20)    | PK  | Unique employee identifier |
| first_name   | VARCHAR(50)    |     | Employee first name        |
| last_name    | VARCHAR(50)    |     | Employee last name         |
| position     | VARCHAR(50)    |     | Job title / role           |
| salary       | DECIMAL(12, 2) |     | Monthly salary (ETB)       |
| email        | VARCHAR(100)   |     | Email address              |
| phone_number | VARCHAR(20)    |     | Contact phone number       |
| hire_date    | DATE           |     | Date of employment         |
| branch_id    | VARCHAR(10)    | FK  | References BRANCH          |

7. LOAN PAYMENTS

| Column       | Type           | Key | Description                          |
|--------------|----------------|-----|--------------------------------------|
| payment_id   | VARCHAR(20)    | PK  | Unique record ID for loan payment    |
| payment_date | DATE           |     | Date payment was processed           |
| amount_paid  | DECIMAL(15, 2) |     | Complete payment currency made (ETB) |

|                   |                |    |                                                |
|-------------------|----------------|----|------------------------------------------------|
| principal_paid    | DECIMAL(15, 2) |    | Portion applied to down principal debt         |
| interest_paid     | DECIMAL(15, 2) |    | Portion applied to interest fees               |
| remaining_balance | DECIMAL(15, 2) |    | Remainder calculation outstanding on loan      |
| employee_id       | VARCHAR(20)    | FK | References EMPLOYEE (Collector processing pay) |
| loan_id           | VARCHAR(20)    | FK | References LOAN                                |

8. USERS

| Column        | Type         | Key | Description                                |
|---------------|--------------|-----|--------------------------------------------|
| user_id       | VARCHAR(20)  | PK  | Core system user account profile ID        |
| username      | VARCHAR(50)  |     | Portal sign-in login name                  |
| password_hash | VARCHAR(255) |     | Encrypted security passphrase string       |
| role          | VARCHAR(20)  |     | Access tier control profile role           |
| status        | VARCHAR(20)  |     | Authorization state (Active/Suspended)     |
| employee_id   | VARCHAR(20)  | FK  | References EMPLOYEE (Nullable if Customer) |
| customer_id   | VARCHAR(15)  | FK  | References CUSTOMER (Nullable if Employee) |
| last_login    | DATETIME     |     | Portal session connection timestamp        |

9. AUDIT LOG

| Column     | Type        | Key | Description                              |
|------------|-------------|-----|------------------------------------------|
| log_id     | VARCHAR(10) | PK  | Unique log operation tracker label       |
| table_name | VARCHAR(50) |     | Target repository data table track point |

|             |              |    |                                                       |
|-------------|--------------|----|-------------------------------------------------------|
| record_id   | VARCHAR(20)  |    | Affected baseline tracking row key identifier         |
| old_values  | TEXT         |    | System record state data before change                |
| new_values  | TEXT         |    | System record state data after change                 |
| action      | VARCHAR(20)  |    | Operational action method flag (INSERT/UPDATE/DELETE) |
| changed_at  | DATETIME     |    | Modification update clock reference stamp             |
| description | VARCHAR(200) |    | Explicit custom narrative comment string              |
| employee_id | VARCHAR(20)  | FK | References EMPLOYEE (Actor of change)                 |

### Summary Result

| Normal Form | Violation Check                                                    | Result |
|-------------|--------------------------------------------------------------------|--------|
| 1NF         | Atomic values, single-valued columns, each row uniquely identified | PASS   |
| 2NF         | No partial dependency on a composite primary key                   | PASS   |
| 3NF         | No transitive dependency through a non-key attribute               | PASS   |
| BCNF        | Every determinant in every FD is a candidate key                   | PASS   |

The Bank Management System database for Gebar Commercial Bank is fully normalized to BCNF. The final updated design eliminates all forms of operational redundancy and modification anomalies while flawlessly preserving all custom functional determinants across nine clean, well-structured relations: Branch, Customer, Account, Loan, Transaction, Employee, Loan Payments, Users, and Audit Log.